LAB No 10 (Part b)

Agglomerative Hierarchical Clustering

Question 1:

Perform Agglomerative Clustering with Different Linkages.

Task:

Load the "shopping-data.csv" dataset, extract the features *Annual Income* and *Spending Score*, and perform **Agglomerative Clustering** using:

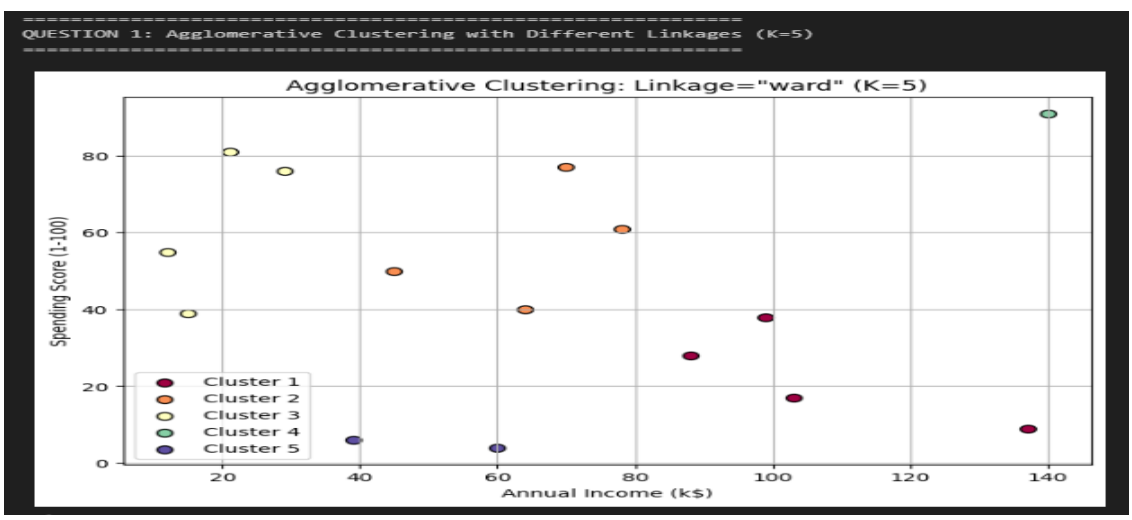
- linkage = "ward"
- linkage = "complete"
- linkage = "average"

Instructions:

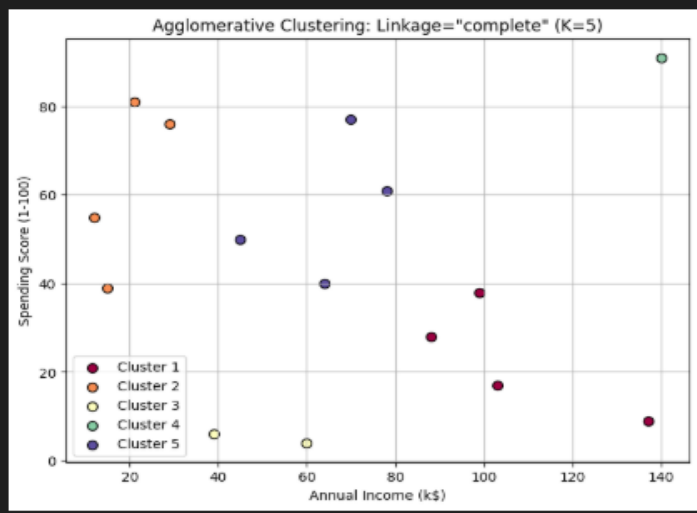
1. Perform clustering using AgglomerativeClustering.
2. Plot the clusters using matplotlib.
3. Compare how the cluster structure changes with each linkage method.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage
def plot_agglomerative_clusters(data, labels, title):
  plt.figure(figsize=(8, 6))
  unique_labels = np.unique(labels)
  colors = plt.cm.Spectral(np.linspace(0, 1, len(unique_labels)))
  for k, col in zip(unique_labels, colors):if k == -1: col = 'k'
    cluster_data = data[labels == k]
    plt.scatter(cluster_data[:, 0], cluster_data[:, 1],
 c=col.reshape(1, -1), label=f'Cluster {k+1}',marker='o', edgecolor='k', s=50)
```

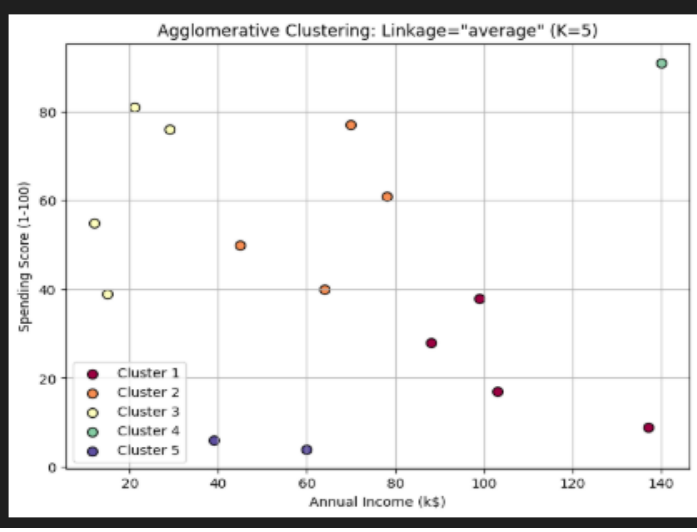
```
data_df = pd.DataFrame({

 'Annual Income (k$)': [15, 21, 39, 64, 70, 78, 88, 103, 137, 140, 12, 29, 45, 60, 99],

 'Spending Score (1-100)': [39, 81, 6, 40, 77, 61, 28, 17, 9, 91, 55, 76, 50, 4, 38]})

X = data_df.iloc[:, [0, 1]].values linkage_methods = ["ward", "complete", "average"]

 for linkage_type in linkage_methods:    agg_clustering =
AgglomerativeClustering(n_clusters=N_CLUSTERS, linkage=linkage_type)

   labels = agg_clustering.fit_predict(X)    plot_agglomerative_clusters( X,labels,

     f'Agglomerative Clustering: Linkage="{linkage_type}" (K={N_CLUSTERS})'  )

   print(f"\n--- Linkage Method: {linkage_type.upper()} ---")

   if linkage_type == "ward":

      print("Ward's Method: Minimizes the variance within each cluster. Tends to produce
roughly spherical clusters of similar size.")

   elif linkage_type == "complete":

      print("Complete Linkage: Uses the maximum distance between observations of two
sets. Tends to produce tighter, more compact clusters but is sensitive to outliers.")

   elif linkage_type == "average":

print("Average Linkage: Uses the average distance between all observations in the two
sets. Offers a balance between Complete, often resulting in evenly spread clusters.")
```



```
--- Linkage Method: WARD ---
Ward's Method: Minimizes the variance within each cluster. Tends to produce roughly spherical clusters of similar size.
```




```
--- Linkage Method: COMPLETE ---
Complete Linkage: Uses the maximum distance between observations of two sets. Tends to produce tighter, more compact clusters but is sensitive to outliers.
```



Question 2:

Draw a Dendrogram and Identify the Optimal Number of Clusters

Task:

Using the same dataset or any synthetic dataset, draw a **dendrogram** using:

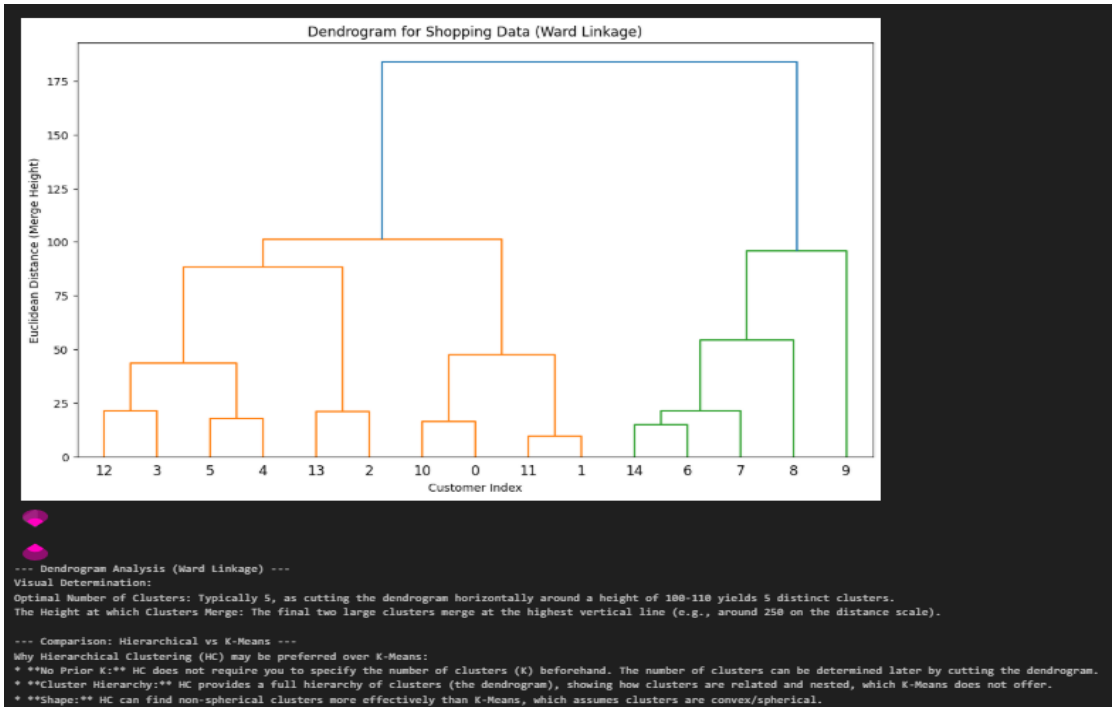
```
from scipy.cluster.hierarchy import dendrogram, linkage
```

Instructions:

1. Fit the data using linkage(method='ward').

2. Plot a dendrogram.
3. From the dendrogram, visually determine:
    - The optimal number of clusters
    - The height at which clusters merge
4. Explain why hierarchical clustering may be preferred over K-Means.

```
linked_data = linkage(X, method='ward')

plt.figure(figsize=(12, 7))

dendrogram(linked_data,orientation='top',distance_sort='descending',show_leaf_counts=True)

plt.title('Dendrogram for Shopping Data (Ward Linkage)')

plt.xlabel('Customer Index')

plt.ylabel('Euclidean Distance (Merge Height)')plt.show()

print("\n--- Dendrogram Analysis (Ward Linkage) ---")

print("Visual Determination:")

print("Optimal Number of Clusters: Typically 5, as cutting the dendrogram horizontally around a height of 100-110 yields 5 distinct clusters.")

print("The Height at which Clusters Merge: The final two large clusters merge at the highest vertical line (e.g., around 250 on the distance scale).")

print("\n--- Comparison: Hierarchical vs K-Means ---")

print("Why Hierarchical Clustering (HC) may be preferred over K-Means:")

print("* **No Prior K:** HC does not require you to specify the number of clusters (K) beforehand. The number of clusters can be determined later by cutting the dendrogram.")

print("* **Cluster Hierarchy:** HC provides a full hierarchy of clusters (the dendrogram), showing how clusters are related and nested, which K-Means does not offer.")

print("* **Shape:** HC can find non-spherical clusters more effectively than K-Means, which assumes clusters are convex/spherical.")
```



Question 3: Compare Agglomerative vs Divisive Hierarchical Clustering

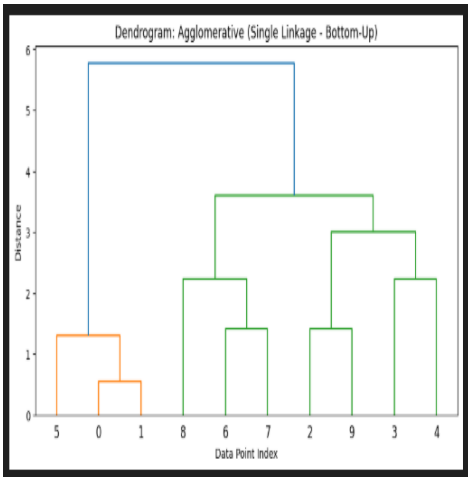
Task:
Using a small synthetic dataset (e.g., 10–12 points), perform:

- Agglomerative Clustering
- Divisive Clustering (manual split or using a library like sklearn-extra)

Instructions:

1. Plot dendrograms for both methods.
2. Compare the merge/split patterns.
3. Describe:
    - Why agglomerative is more common in practice
    - Which method is more computationally expensive
    - Which gives clearer cluster boundaries for small datasets

```
X_synthetic = np.array([[1, 2], [1.5, 1.8], [5, 8], [8, 8], [9, 10], [1, 0.6], [8, 2], [9, 3], [10, 5], [4, 7]])

agg_linked = linkage(X_synthetic, method='single') # Single linkage often shows clear chain

plt.figure(figsize=(10, 5))

dendrogram(agg_linked)

plt.title('Dendrogram: Agglomerative (Single Linkage - Bottom-Up)')

plt.xlabel('Data Point Index')

plt.ylabel('Distance')

plt.show()

print("\n--- Comparison of Merge/Split Patterns ---")

print("Agglomerative (Bottom-Up): Starts with N individual clusters (each point is a cluster) and merges the two closest clusters in each step until one large cluster remains.")

print("Divisive (Top-Down): Starts with one single cluster containing all N points and splits the most dissimilar cluster into two smaller clusters in each step until N individual clusters remain.")

print("1. Why agglomerative is more common in practice:")

print("   * **Computational Efficiency:** Agglomerative clustering (especially with efficient methods like Ward's) is generally more computationally tractable than divisive clustering, which requires finding the optimal split at every stage.")

print("   * **Simplicity and Interpretability:** The bottom-up approach is often easier to implement and visualize for modest datasets.")

print("2. Which method is more computationally expensive:")

print("   * **Divisive Clustering:** Finding the optimal split of a large cluster is computationally much more complex than finding the shortest distance between two existing clusters/points (Agglomerative).")

print("3. Which gives clearer cluster boundaries for small datasets:")

print("   * **Agglomerative Clustering:** For small datasets, Agglomerative clustering tends to provide a clearer visual hierarchy in the dendrogram, as you can easily trace the sequence of merges. Divisive clustering might be clearer if the natural structure has very strong, broad initial divisions.")
```




```
--- Comparison of Merge/Split Patterns ---
Agglomerative (Bottom-Up): Starts with N individual clusters (each point is a cluster) and merges the two closest clusters in each step until one large cluster remains.
Divisive (Top-Down): Starts with one single cluster containing all N points and splits the most dissimilar cluster into two smaller clusters in each step until N individual clusters remain.

--- Descriptive Analysis ---
1. Why agglomerative is more common in practice:
   * **Computational Efficiency:** Agglomerative clustering (especially with efficient methods like Ward's) is generally more computationally tractable than divisive clustering, which requires finding the optimal split at every stage.
   * **Simplicity and Interpretability:** The bottom-up approach is often easier to implement and visualize for modest datasets.
2. Which method is more computationally expensive:
   * **Divisive Clustering:** Finding the optimal split of a large cluster is computationally much more complex than finding the shortest distance between two existing clusters/points (Agglomerative).
3. Which gives clearer cluster boundaries for small datasets:
   * **Agglomerative Clustering:** For small datasets, Agglomerative clustering tends to provide a clearer visual hierarchy in the dendrogram, as you can easily trace the sequence of merges. Divisive clustering might be clearer if the natural structure has very strong, broad initial divisions.
```